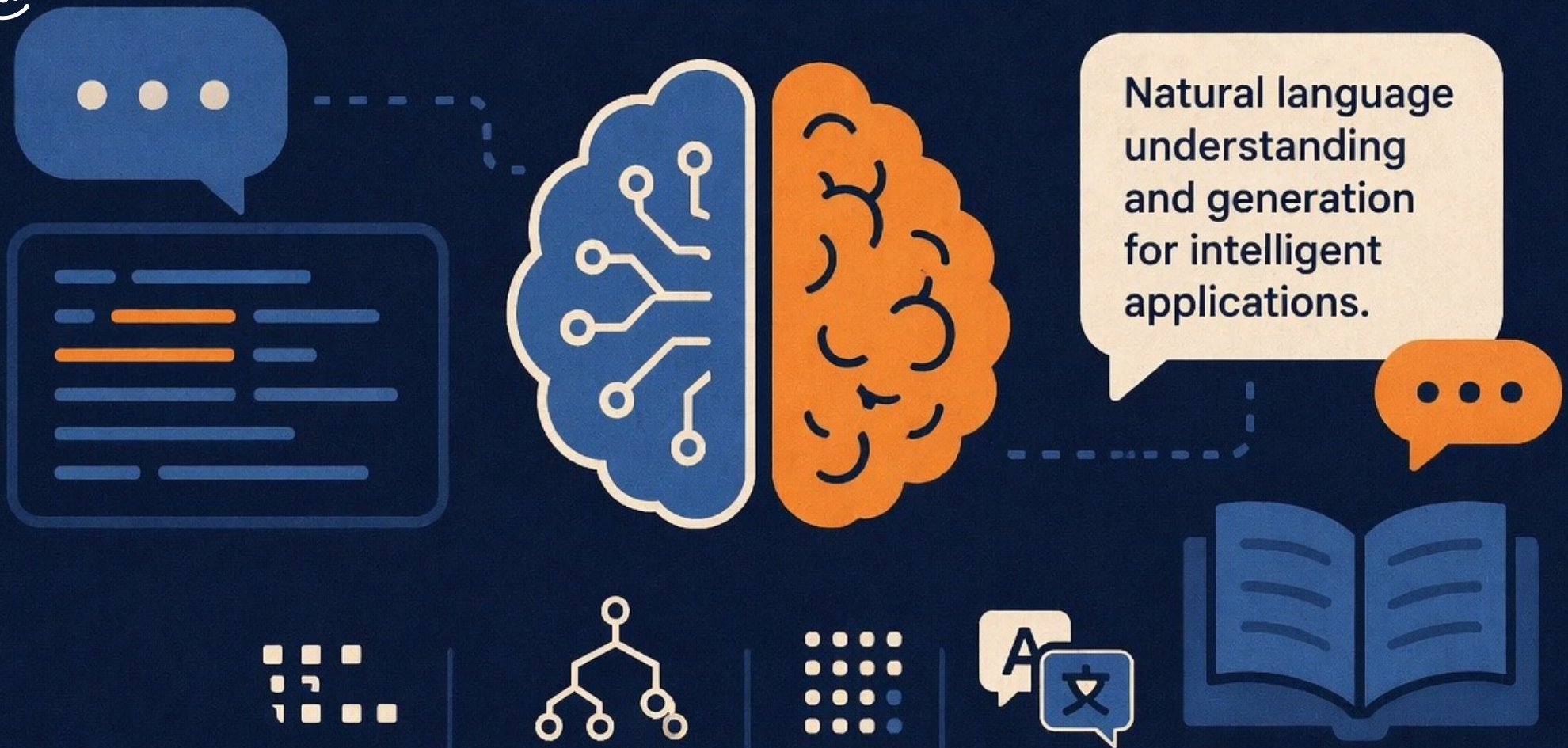
# NATURAL LANGUAGE PROCESSING

class starts @ 9:05 pm



# Lecture: NER

↳ Named Entity Recognition

#Agenda

1) Bi-directional LSTM
2) GRU
3) NER
4) Bi-directional LSTM + CRF
5) Coding

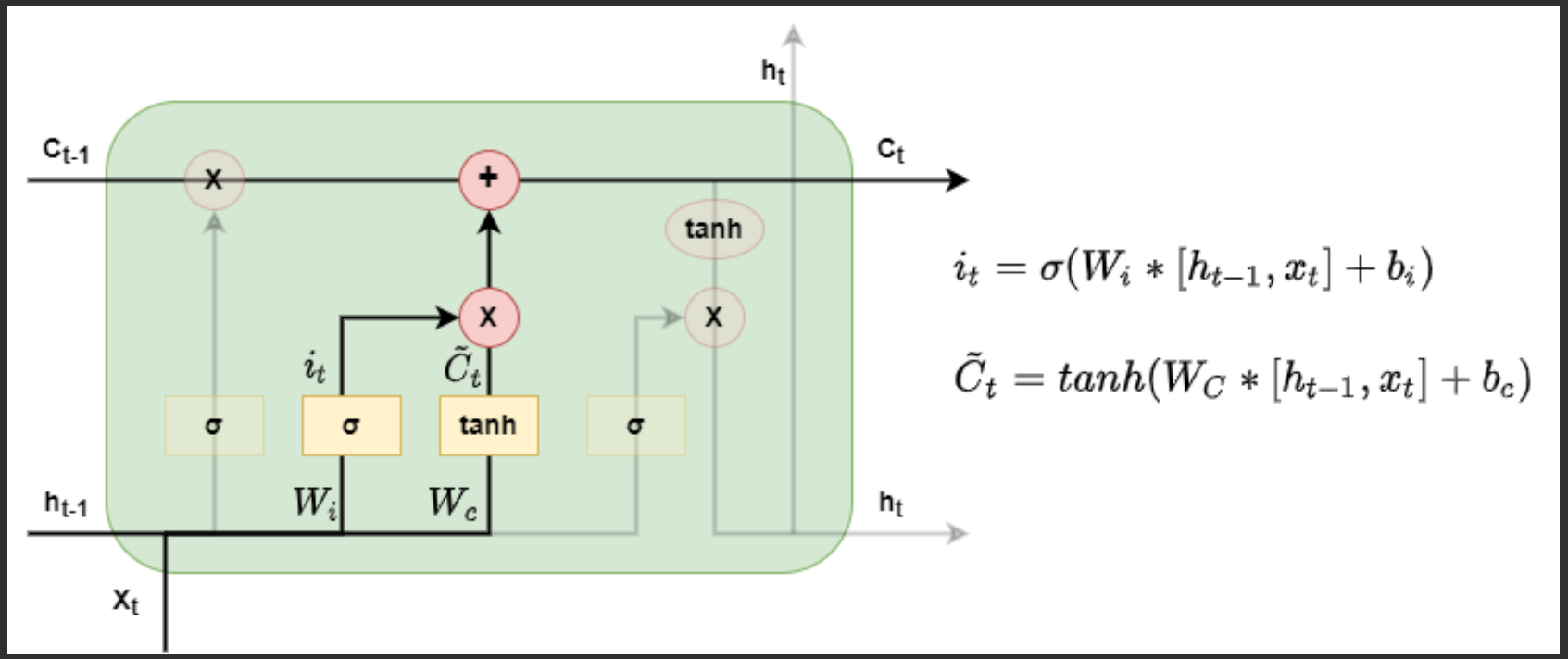
# Recap

LSTM → Long Short Term memory

↳ Cell State (Long term memory)

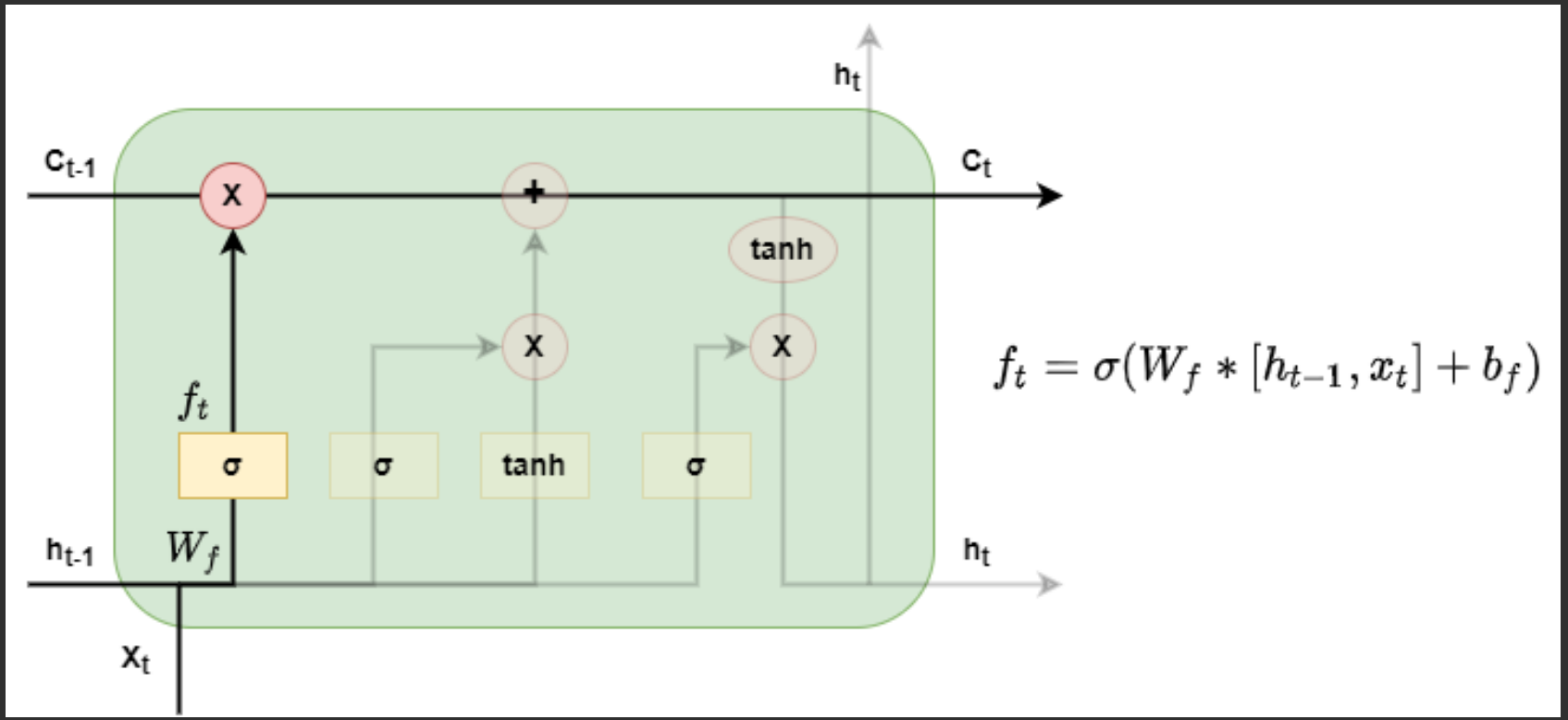
Hidden State (Short term memory)

"Long Term dependency"

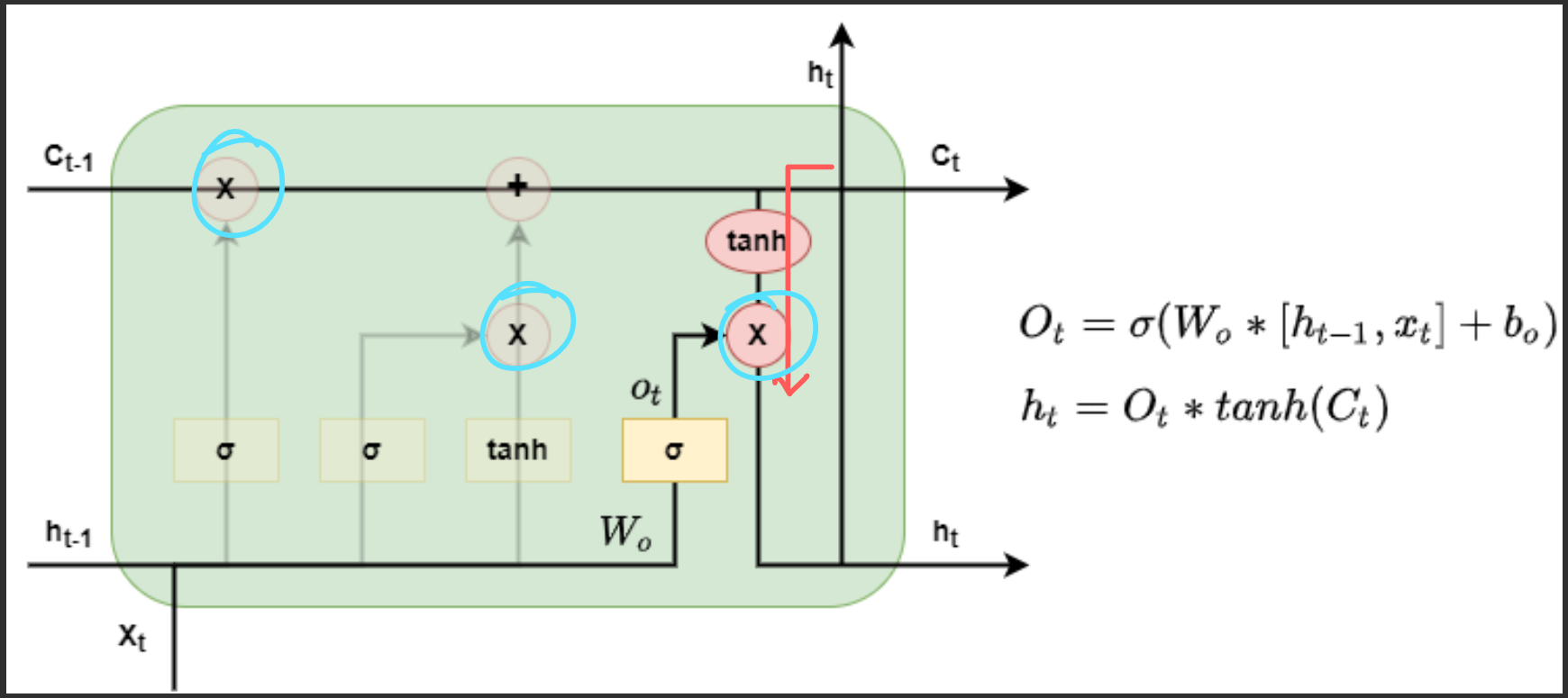



$$i_t = \sigma(W_i * [h_{t-1}, x_t] + b_i)$$

$$\tilde{C}_t = tanh(W_C * [h_{t-1}, x_t] + b_c)$$



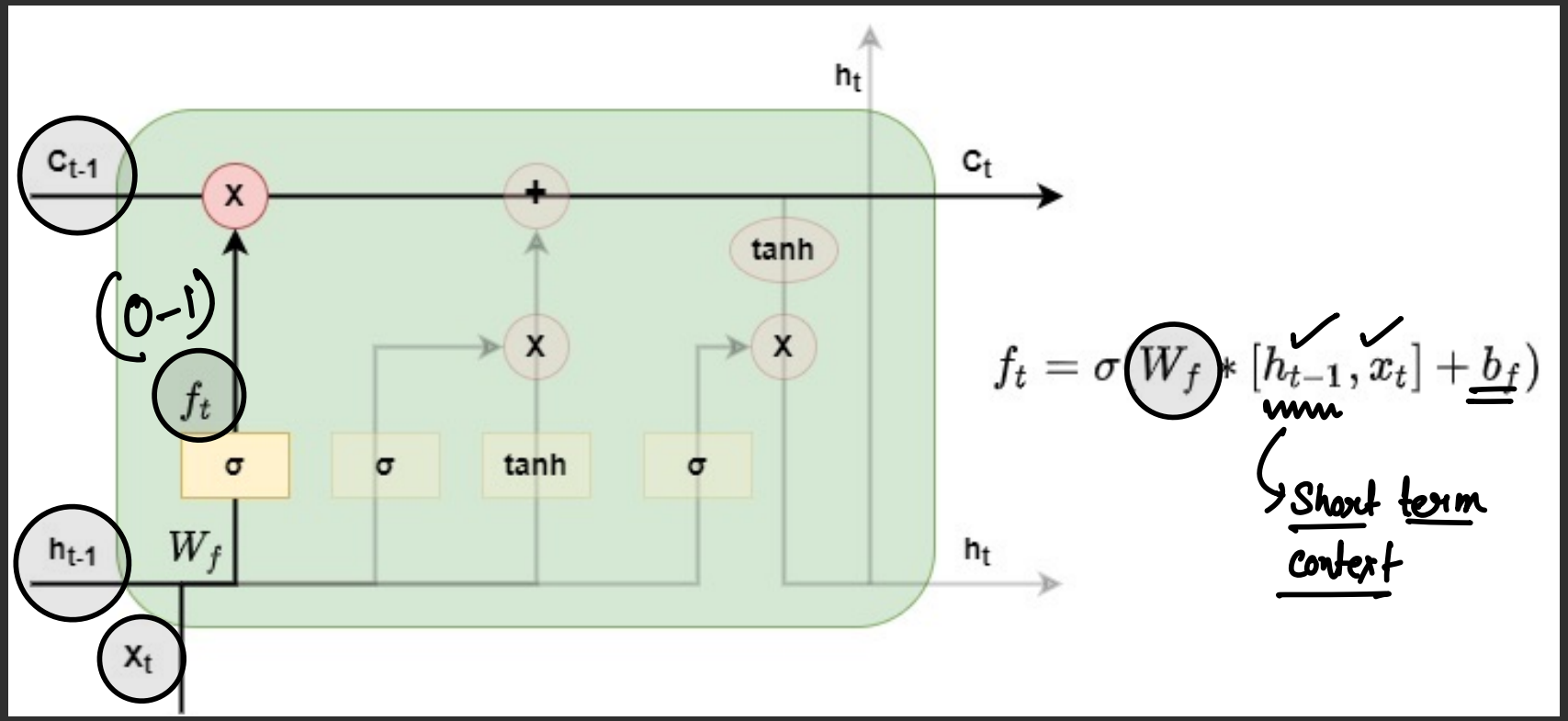

$$f_t = \sigma(W_f * [h_{t-1}, x_t] + b_f)$$



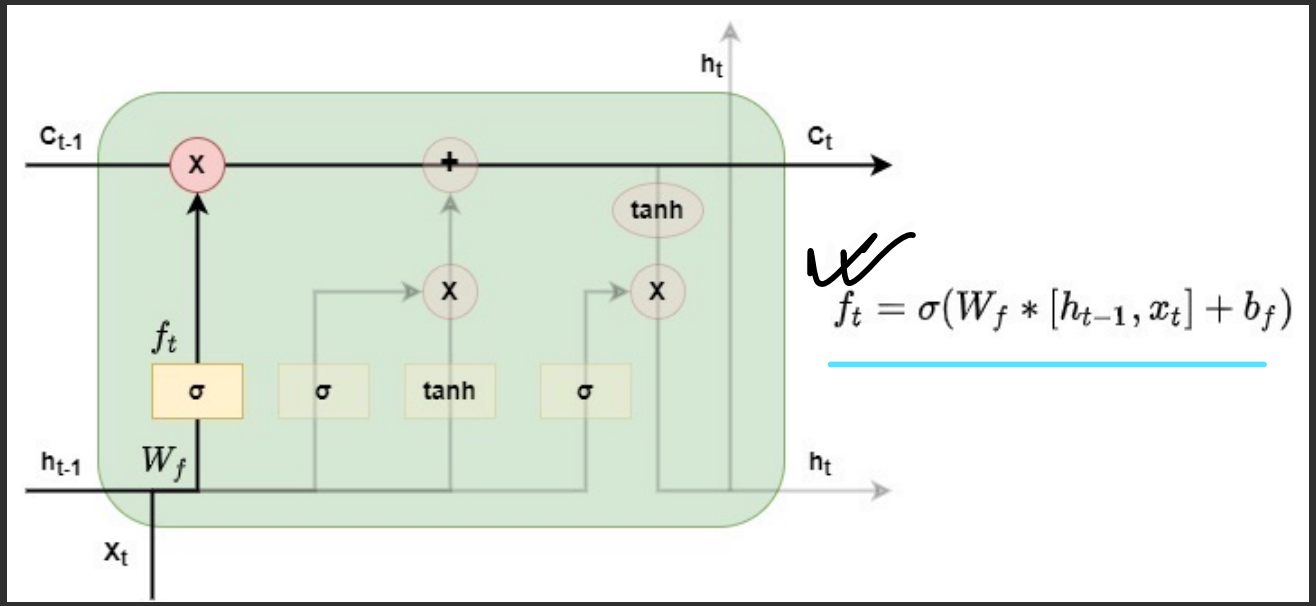

$$O_t = \sigma(W_o * [h_{t-1}, x_t] + b_o)$$

$$h_t = O_t * tanh(C_t)$$

[0, 1, 0, 1] + 1 ⇒ [1, 2, 1, 2]




I went to paris last year & found that place to be akaish, This I'll go to Egypt




$W_f, b_f$

↳ Random init

$f_t = [0.68, 0.55]$

↳ output of forget gate

I love NLP

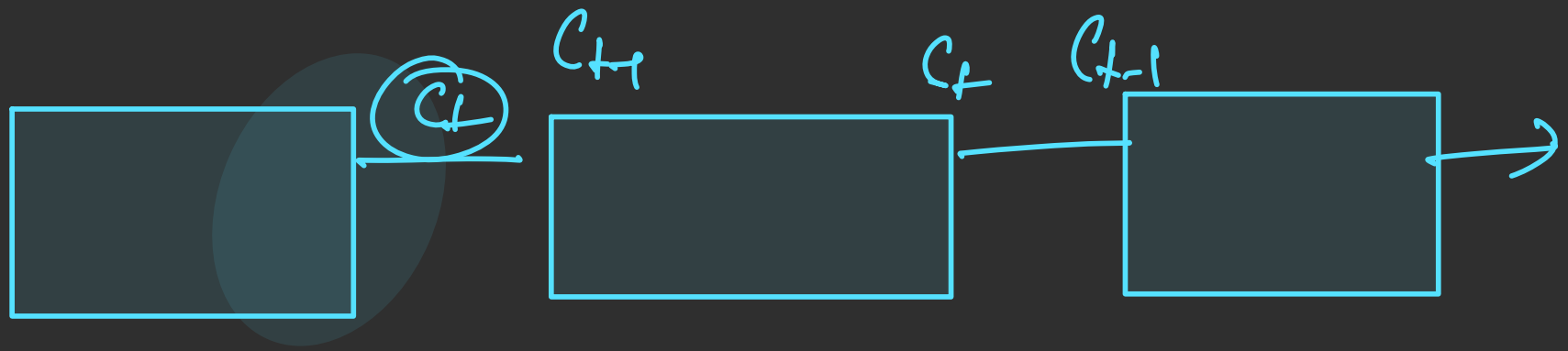
love = [0.5, 0.1]

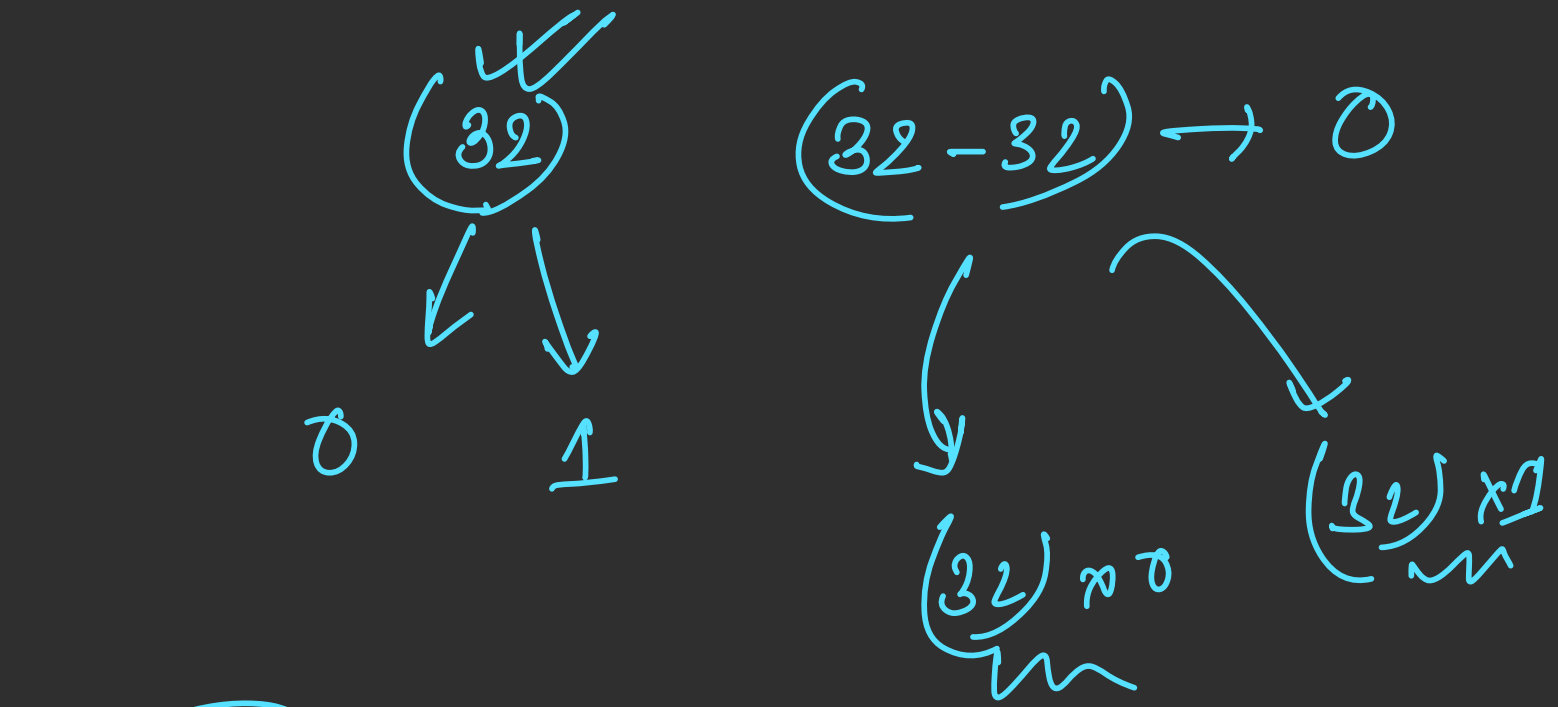
I = [0.6, −0.2]

$x_t$

$C_{t-1} \Rightarrow [0.6, -0.2]$

$f_t \cdot C_{t-1} = [0.46, -0.23]$


$C_{t+1}$
$C_t$
$C_{t-1}$
$C_t$




$W \cdot x$

weights

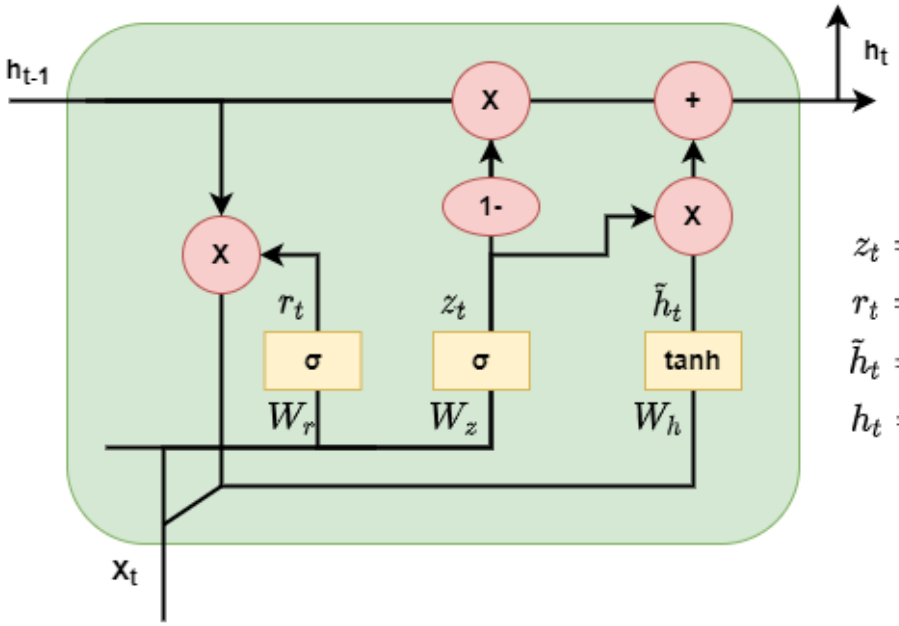
$W + x$

$Word_1$ (Jack) $\longrightarrow$ $W_f, W_i, W_c, W_o$

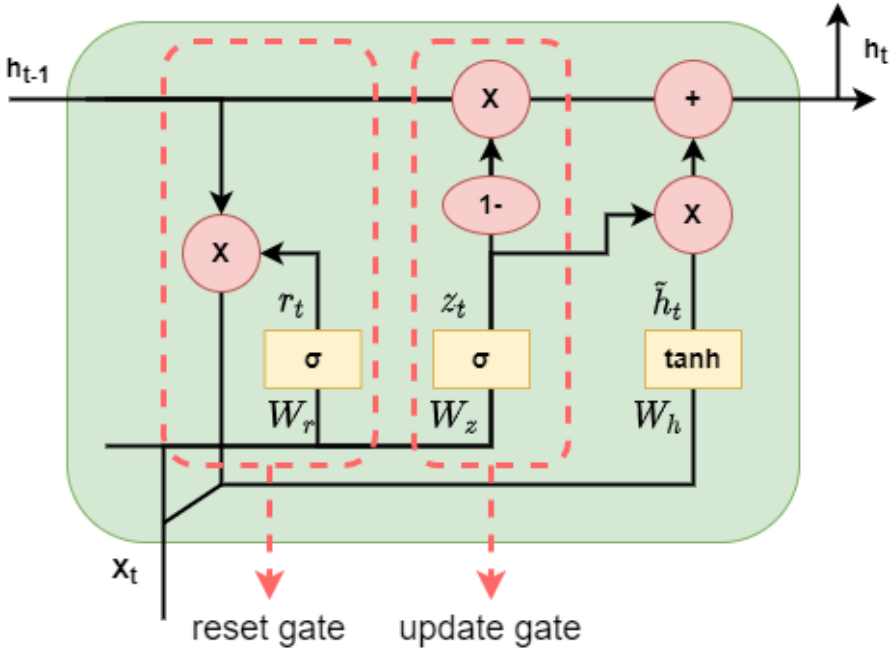
$Word_2$ (went) $\longrightarrow$ $W_f, W_i, W_c, W_o$

"Drawback of LSTM" "Hard to train"

# Variants of LSTM - GRU → Gated Recurrent Unit




$$z_t = \sigma(W_z * [h_{t-1}, x_t])$$
$$r_t = \sigma(W_r * [h_{t-1}, x_t])$$
$$\tilde{h}_t = tanh(W_h * [h_{t-1}, x_t])$$
$$h_t = (1 - z_t) * h_{t-1} + z_t * \tilde{h}_t$$



Another Variant of LSTM is the GRU

1. It has 3 weight matrices Wr,Wz,Wh which is 1 weight matrix less than LSTM
2. There is only one hidden state ht in GRU unlike a separate cell state and hidden state in LSTM.

It has 2 gates a reset gate and a update gate.
The update gate works similar to forget gate in LSTM as it determines what information to forget and what new information to update.
The reset gate determines how much past information to forget

# Difference b/w LSTM & GRU

## LSTM

3 gates + 1 memory cell

Forget gate → decides what to remove from memory

Input gate → decides what new info to add

Output Gate → decides what to output

Cell state → long term memory

# GRU (Gated Recurrent Unit)

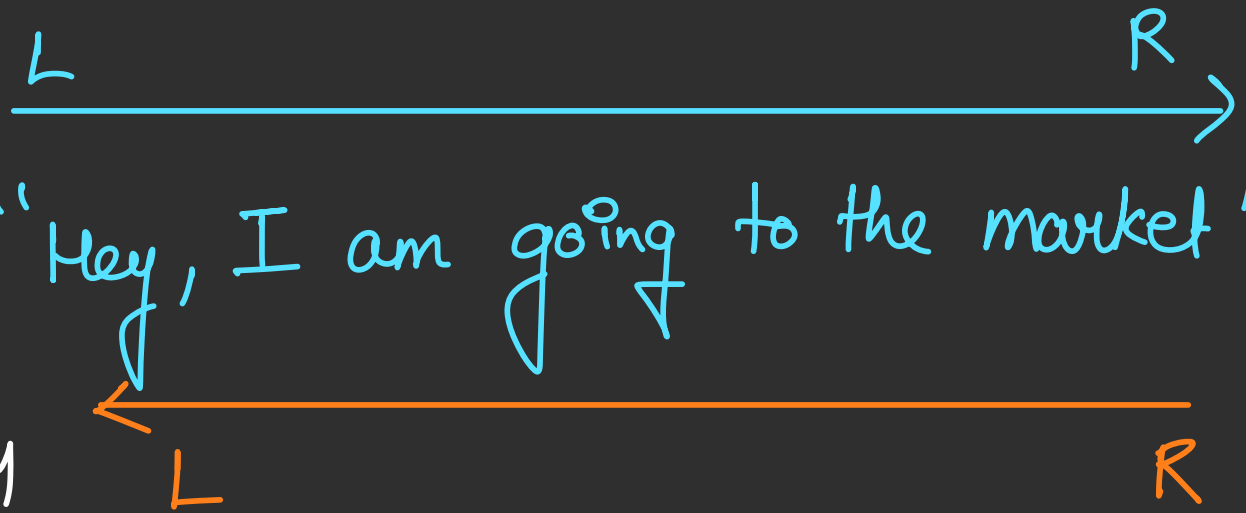
Has 2 gates only

Update gate → combines forget + input

Reset gate → controls past information usage

No separate cell state ($C_t$) → uses hidden state directly.

# Bi-directional LSTM / GRU

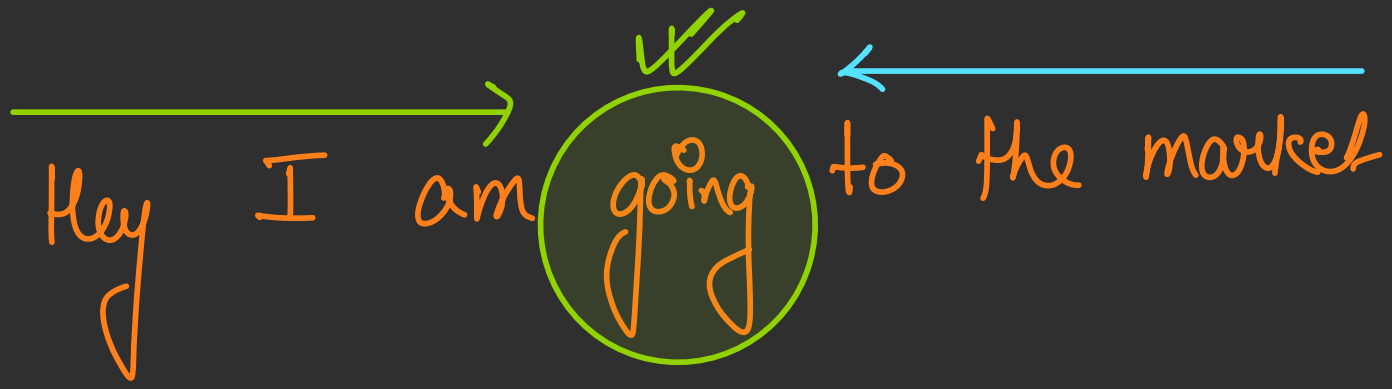



L LSTM

1) Hey
I
am
⋮

R LSTM

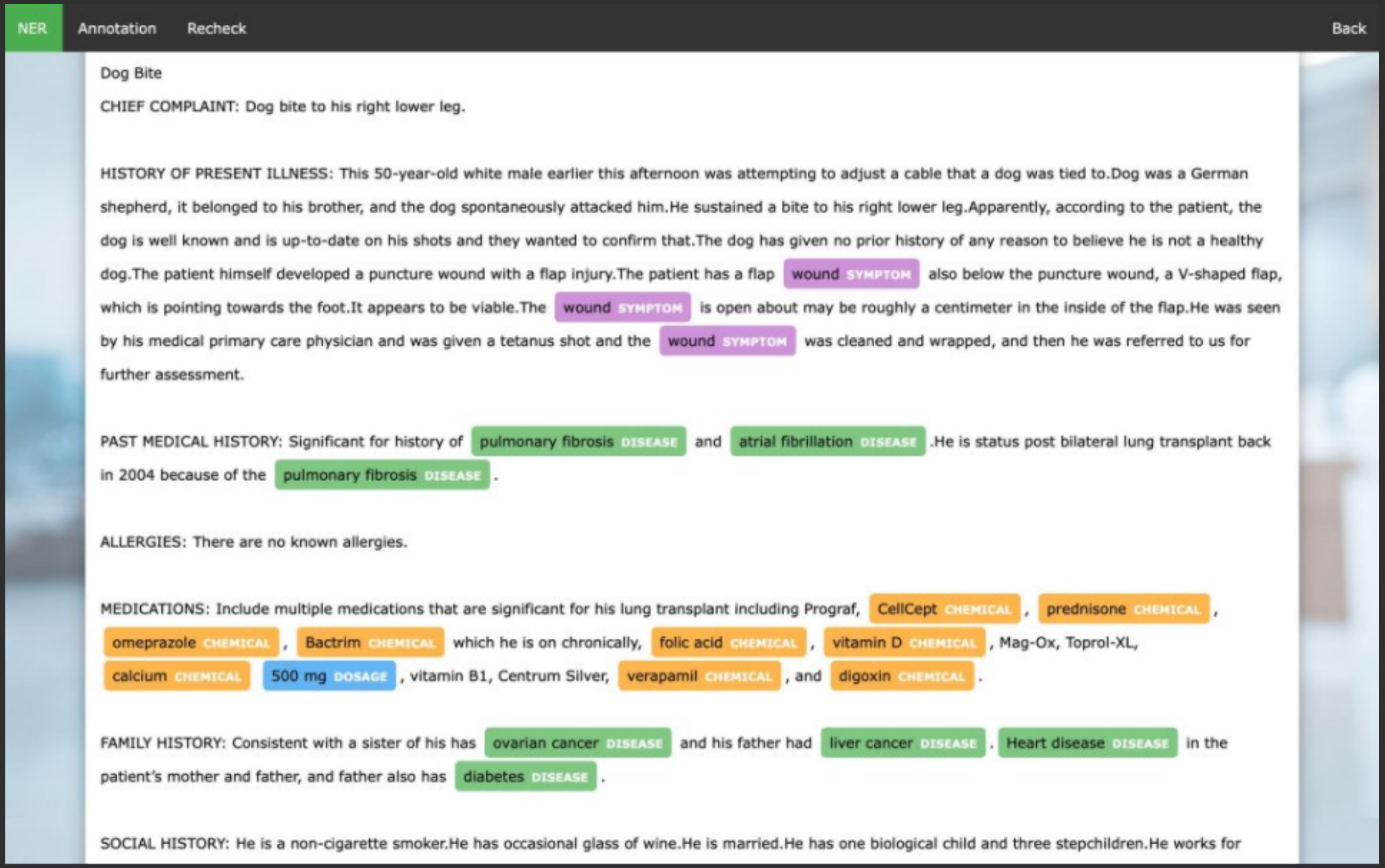
1) market
2) the

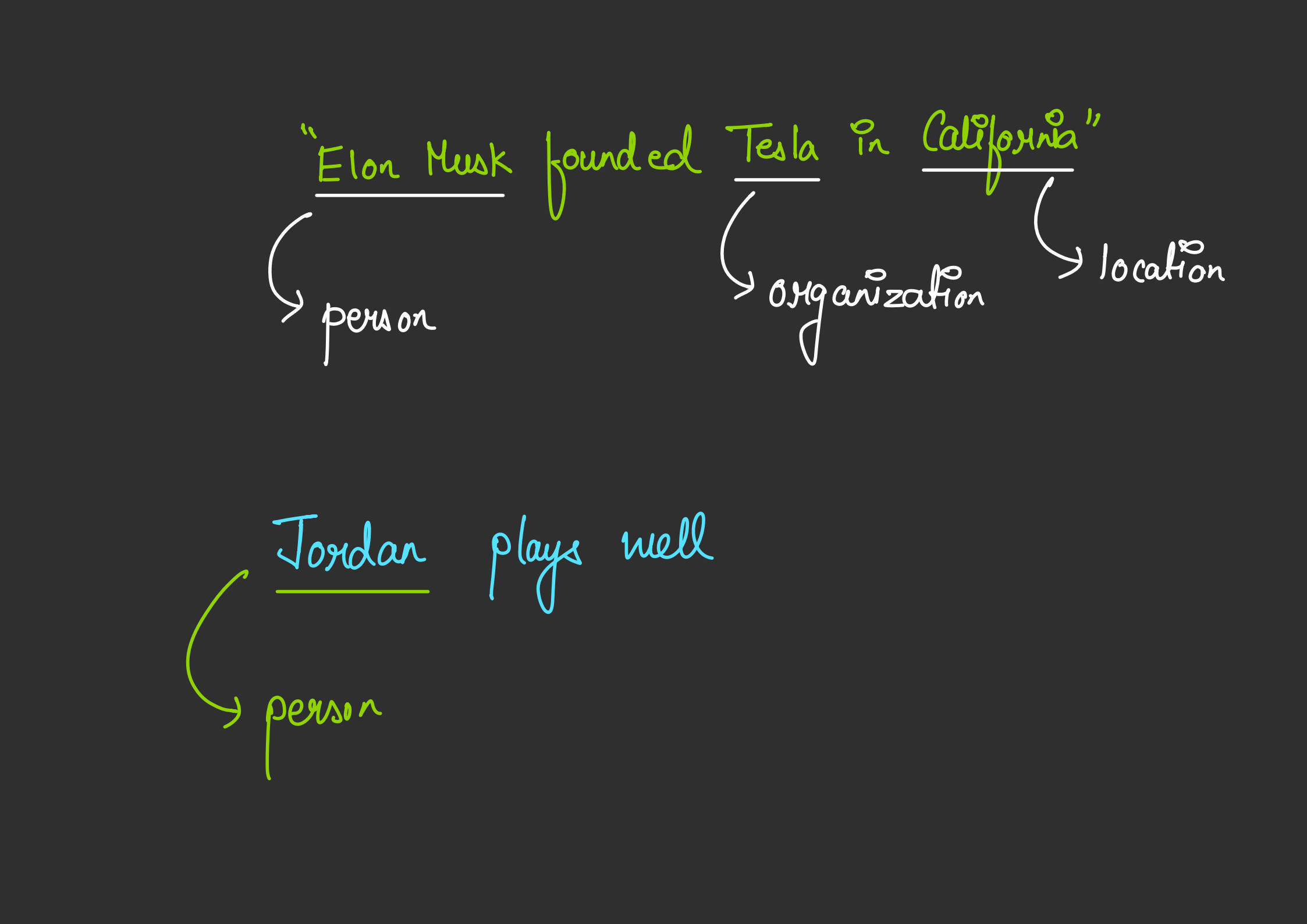



L LSTM

R LSTM

# What is NER?


NER
Annotation
Recheck
Back
Dog Bite
CHIEF COMPLAINT: Dog bite to his right lower leg.
HISTORY OF PRESENT ILLNESS: This 50-year-old white male earlier this afternoon was attempting to adjust a cable that a dog was tied to.Dog was a German shepherd, it belonged to his brother, and the dog spontaneously attacked him.He sustained a bite to his right lower leg.Apparently, according to the patient, the dog is well known and is up-to-date on his shots and they wanted to confirm that.The dog has given no prior history of any reason to believe he is not a healthy dog.The patient himself developed a puncture wound with a flap injury.The patient has a flap wound SYMPTOM also below the puncture wound, a V-shaped flap, which is pointing towards the foot.It appears to be viable.The wound SYMPTOM is open about may be roughly a centimeter in the inside of the flap.He was seen by his medical primary care physician and was given a tetanus shot and the wound SYMPTOM was cleaned and wrapped, and then he was referred to us for further assessment.
PAST MEDICAL HISTORY: Significant for history of pulmonary fibrosis DISEASE and atrial fibrillation DISEASE .He is status post bilateral lung transplant back in 2004 because of the pulmonary fibrosis DISEASE .
ALLERGIES: There are no known allergies.
MEDICATIONS: Include multiple medications that are significant for his lung transplant including Prograf, CellCept CHEMICAL , prednisone CHEMICAL , omeprazole CHEMICAL , Bactrim CHEMICAL which he is on chronically, folic acid CHEMICAL , vitamin D CHEMICAL , Mag-Ox, Toprol-XL, calcium CHEMICAL 500 mg DOSAGE , vitamin B1, Centrum Silver, verapamil CHEMICAL , and digoxin CHEMICAL .
FAMILY HISTORY: Consistent with a sister of his has ovarian cancer DISEASE and his father had liver cancer DISEASE . Heart disease DISEASE in the patient's mother and father, and father also has diabetes DISEASE .
SOCIAL HISTORY: He is a non-cigarette smoker.He has occasional glass of wine.He is married.He has one biological child and three stepchildren.He works for

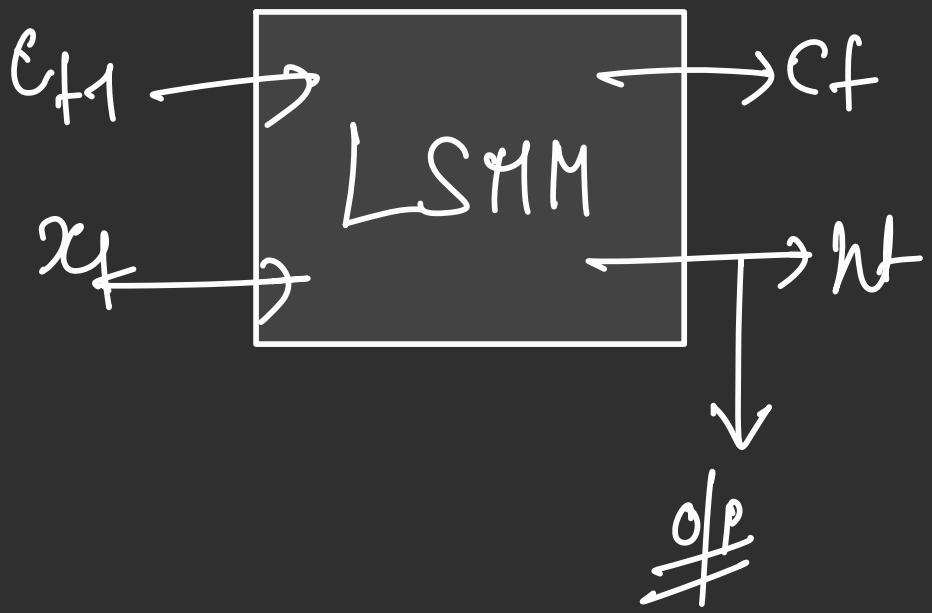

"Elon Musk founded Tesla in California"
person
organization
location
Jordan plays well
person

Elon Musk is the owner of Tesla

Entity

Elon Musk

"Bi-directional LSTM"



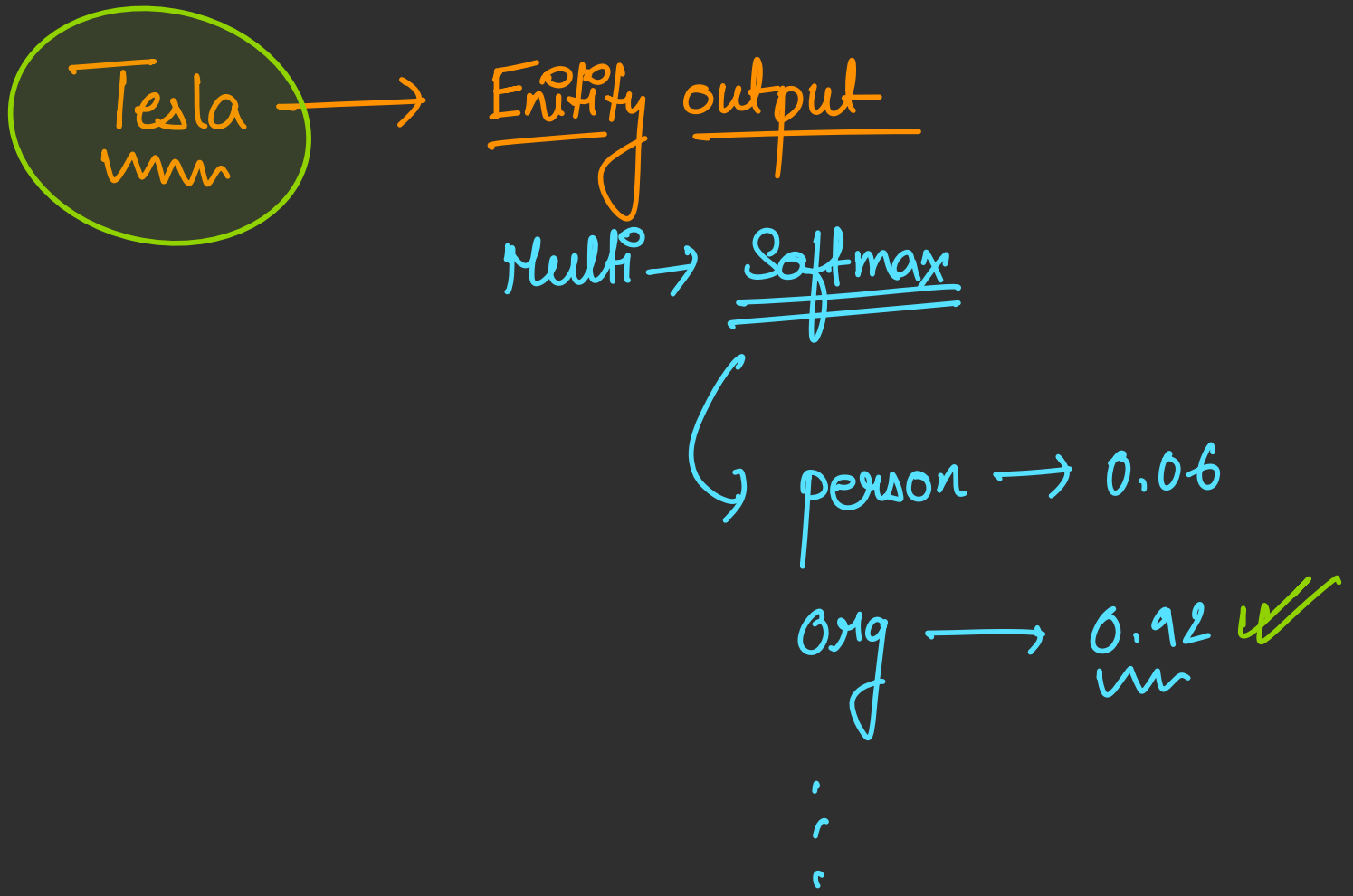
Named-entity recognition (NER), in general, (also known as entity identification or entity extraction) is a subtask of information extraction (text analytics) that aims at finding and categorizing specific entities.

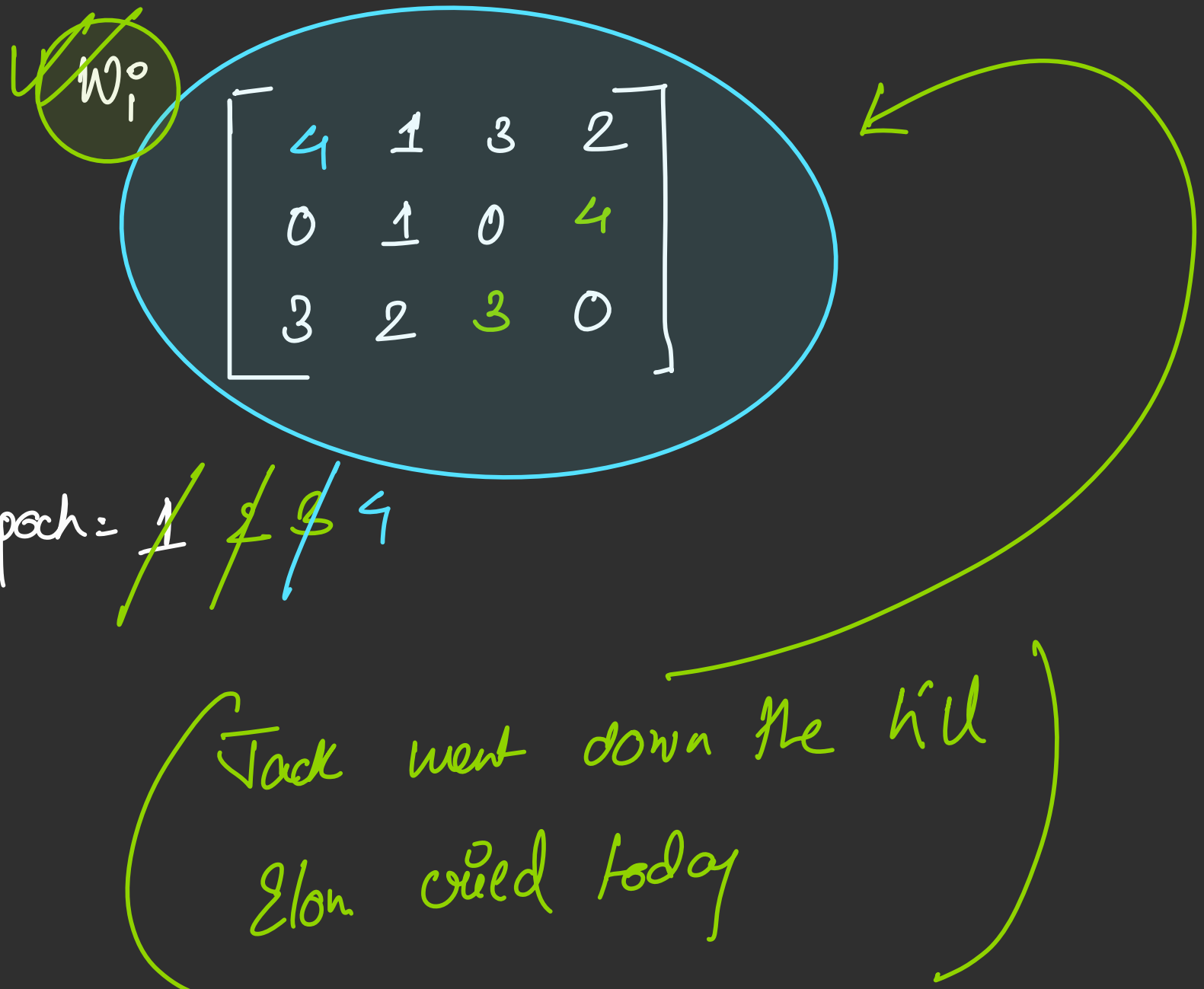
NER involves the identification of proper names in texts, and the classification of these names into a set of predefined categories of interest like person, location, organization, drug, time, etc.

It refers to extracting ‘named-entities’ from text. Named-entities denotes to words in a sentence representing real-world objects with proper names like:

Person’s name (Ramu, Raja, Seeta, etc.),
Countries (India, Sri Lanka, etc),
Organization (Google, Facebook, etc.)
or anything that has been given a specific name.

| Type | Tag | Sample Categories | Example sentences |
|---|---|---|---|
| People | PER | people, characters | **Turing** is a giant of computer science. |
| Organization | ORG | companies, sports teams | The **IPCC** warned about the cyclone. |
| Location | LOC | regions, mountains, seas | The **Mt. Sanitas** loop is in **Sunshine Canyon**. |
| Geo-Political Entity | GPE | countries, states, provinces | **Palo Alto** is raising the fees for parking. |
| Facility | FAC | bridges, buildings, airports | Consider the **Golden Gate Bridge**. |
| Vehicles | VEH | planes, trains, automobiles | It was a classic **Ford Falcon**. |


Tesla
Entity output
Multi → Softmax
person → 0.06
org → 0.92


$W_1^0$
4 1 3 2
0 1 0 4
3 2 3 0
Epoch: 1 4
Jack went down the hill
Elon cüld today